Anomalous Returns, Risk Premiums and Diversification: Evidence from Frontier Market

Hilal Anwar Butt*

*Assistant Professor and Research Fellow at Department of Economics and Finance, Institute of Business Administration Karachi, Pakistan.

Contact # 00-92-03365213803

Email: habutt@iba.edu.pk

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A simple size and volatility based zero-investment strategies yield 30% to 50% annual returns in Karachi stock exchange (KSE), Pakistan. These returns are quite higher in comparison to comparable evidence for the most efficient market of the US. This study indicates that the higher returns as such is not a vindication of market inefficiency rather, it is a compensation to investors for being exposed to market and illiquidity related local risks. Further we find that KSE provide significant portfolio diversification opportunities to the international investors, even for those investors who are just interested in bigger and the most liquid stocks.

JEL Codes: G10, G12, G15

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1 Introduction

The challenge for efficient market hypothesis is the existence of predictable anomalous returns, which can be generated by taking long and short positions on some stocks on the basis of their publically available characteristics. In the theory of efficient market, such predictable anomalous returns must be linked with the predictability of risk premium. So, that the notion of higher risk and higher gain is reconciled. This reconciliation is generally established through asset pricing models, like capital asset pricing model (CAPM) of Sharpe (1964), Linter (1965) and Mossin (1966), three factor model of Fama and French (1993) and four factor model of Carhart (1997).

The recent addition in this league of models is proposed by Hou, Xue and Zhang (2014) and Fama and French (2015) five factor model with the realization that the investment and operating profitability also constitute market wide risks. These aforesaid models are tested extensively to explain the anomalous returns for the US market in particular, and for other markets in general. There are two issues involved to implement such models for the most of emerging/frontier markets. Firstly the data for firm related characteristics is not adequately available to construct well diversified market measures of risk, secondly the effect of illiquidity is not explicitly accounted for in these models. Therefore, to study the anomalous returns for the frontier market like Pakistan the liquidity adjusted capital asset pricing model (LCAMP) of Acharya and Pedersen (2005) is an appropriate choice. As within the framework of LCAPM one can capture the pricing effect of level of illiquidity, market and illiquidity risks.

The selection of Karachi stock exchange (KSE), as emerging/frontier market is made to compare the magnitude of anomalous returns available in less researched market in comparison to the US market on the basis of commonly known strategies. Secondly to analyze if these returns can be rationalized within the risk and return framework offered by some model by implying effect of local illiquidity and market risk. Lastly, the returns that are offered by KSE provide any diversification benefit to international investors.

We find that anomalies are of very high magnitude for the data of the period of May 1993-June 2015 (1993-2015 onwards), the excess return by being short on 20% highly capitalized stocks and long on 20% minimum capitalized stocks is 31.57% on annual basis. Similarly portfolios based on monthly idiosyncratic volatility of returns (volatility) have the return dispersion of 50.43% on annual basis between the 20% the most volatile and 20% the least volatile portfolios. Whereas, for the same time period the zero-investment strategy for the US ,yields annual excess return of 4.72%[1] for ten equally weighted size portfolios, the same hold true of volatility related portfolios.

Although the size and volatility related anomalies have high magnitude, however within the framework of Acharya and Pedersen (2005), LCAPM these higher returns are linked with premium associated with level of illiquidity, market and illiquidity risks. For instance, 85% of anomalous returns on size portfolios are linked with illiquidity level and market risk. Another extension of model that incorporate illiquidity level and illiquidity risk almost explain 73% of the excess returns. Similarly for volatility related portfolios, the 100% of excess return on the most volatile

[1] The analysis are conducted by extracting the portfolio related information for the US market from Fama and French data library.

and least volatile portfolios is explained by market and illiquidity risk. Whereas, the level of illiquidity is not important for the volatility related portfolios. This is quite understandable as level of illiquidity is linked with the returns on size related portfolios but the same does not hold for volatility related portfolios. These results are consistent with the previous studies of Bekaert et al. (2007) and Lee (2011), in which local illiquidity risk is found to be quite relevant for pricing in emerging markets. However these studies are panel based, therefore the validity of the results cannot be generalized for all countries. For instance, in Lee (2011) the illiquidity premium for emerging markets is 5.58%[2] on annual basis by implying local illiquidity risk in LCAPM. In our study the premium associated with illiquidity risk are as high as 17.26% and 59.33% for size and volatility related portfolios after controlling of level of illiquidity.

Lastly, this study also address the question like, does the frontier/emerging market offer any diversification benefit to international investors. We find that for the US investor the diversification effect is economically quite significant. Once the risk of the US investor is gauged through three US specific, Fama and French (1993) factors, the annual anomalous returns of 29.16% and 43.20% on size and volatility strategy remain unexplained. Of course the underlying assumption of these results are that KSE is fully liberalized market, which is quite far-fetched. Once we implied the excess return on those firms listed on KSE (the most liquid and large firms) which are included in S&P's extended frontier 150 index, the yearly excess return of 19.68% is not explained by the three factors Fama and French (1993) model. These results indicate that KSE offers economically meaningful diversification opportunities to the foreign investors.

[2] The illiquidity premiums are discussed in section 5.1 Empirical results for local liquidity risk of Lee (2011).

The rest of the paper is organized as, the section 2 discusses the KSE and illiquidity related issues, in section 3 we construct test portfolios and elaborate its characteristics, section 4 illustrates the methodology and empirical analysis and section 5 concludes.

2. Karachi Stock Market and Illiquidity.

The data for the analysis is downloaded through DataStream for the period of 1993-2015. In the initial screening of the data all non-common stocks are deleted. However the dead firms are retained to avoid survivorship problem. There are some other cleaning procedures that are adopted to clean the data of the recording errors for DataStream database which is indicated in previous research. For instance following Griffin et al. (2010) and Ince and Porter (2006), daily returns are set to be missing if they increase and decrease significantly such that, $r_{t-1} > 100\%$ or $r_t > 100\%$ and $(1 + r_t) * (1 + r_{t-1}) - 1 \leq 50\%$. In addition to this criteria we show a daily return missing, if it is greater than 200%. For monthly returns as well, we set those monthly return to be zero, if they increase and then revert such that they satisfy this condition $r_{t-1} > 300\%$ or $r_t > 300\%$ and $(1 + r_t) * (1 + r_{t-1}) - 1 \leq 50\%$. , Lastly we show all monthly returns that are greater than 800% to be missing.

After all cleaning the total coverage of firms including the dead firms is 421 for the period of 1993-2015. The average number of stocks in KSE is 229, however there is considerable number of the stocks which are traded for maximum of three days within a month. If such stocks are excluded then the average number of stocks reduced to 139. The Figure 1, traces the percentage of such firms in the sample for the period of 1993-2015. It is quite visible that concentration of such firms has decreased substantially over the time. The portion of such firms is lesser than 15% for the last five years, which is significantly lower in

[Insert Figure 1 here]

comparison to initial years. This indicates the increased liquidity of KSE market. To add to this evidence of increased liquidity, the liquidity measure for the stocks listed in KSE market is estimated as the ratio of monthly zero returns over total trading days in a month. This is expressed as under,

$$ZR = ZRD_{i,t} / TD_{i,t} \tag{1}$$

Such that, $ZRD_{i,t}$, is total zero returns in a month for a stock and, $TD_{i,t}$, shows the total trading days in any month. This liquidity measure is extensively been used in the literature such as by Bekaert et al. (2007) and Lee (2011) etc. The market illiquidity is simply the average of all firms *ZR* measure and it is shown in Figure 2, there is a straightforward exhibition of the effect that number of the firms that are traded is increased over the time. This increased trade can be attributed to decreasing transaction cost/illiquidity of the KSE market.

[Insert Figure 2 here]

In Figure 2, there is a visible hump around December 2008, and that is owed to the imposition of "floor rule" in the context of financial crises. This results in practical shutdown of the KSE market, which led to exit of MCSI Pakistan index from MCSI emerging market index. However, condition improved in terms of tradability of firms afterwards which is also visible in Figure 2. Recently

KSE is described as the best hidden frontier market[3], with the 16% growth for last 12 months making it among the top ten the best performing markets. Resultantly the inclusion of KSE in emerging market index provided by MSCI is likely to be under its review agenda for the year 2016.

3. Test Portfolios for KSE

There are different characteristics reported in the literature that are linked with the returns around different markets. Of them size and volatility of stocks returns are chosen for their relevance for small sized emerging/frontier market. As illiquidity, which is usually related with size and volatility are the characteristics that matter the most for the investors in such markets. There is one additional benefit of choosing these characteristics such that, level of illiquidity is linked with size and same does not hold for volatility, this point is highlighted in coming paragraphs. Nevertheless, there is significant variation in returns of the stocks based upon their apparently dissimilar level of illiquidity.

As the average number of stocks in KSE is 139, therefore only 5 portfolios are generated using each characteristic. Using size[4] based information of the month of January 1993, the returns for the month of March 1993[5] are allocated to five portfolios. Such that portfolio S-1 is the collection of those stocks whose one month's preceding size is less than or equal to 20% percentile of the size of all available firms. Similarly the portfolios S-2, S-3, S-4 and S-5 are the collection of those stocks whose preceding month's size are increasing monotonically by 20%. We adopt for monthly sorting procedure to incorporate the maximum possible information at firm level into the returns[6].

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[3] Bloomberg date June 30, 2015. Link http://www.bloomberg.com/news/articles/2015-06-30/in-best-hidden-frontier-market-boom-signals-pakistan-revival.
[4] Which is number of shares outstanding multiplied by the end of month prices of the firm.
[5] One month is left-out to control for the short-term reversal effect.
[6] A rationale of such strategy is elucidated in foot note 21 of Sadka (2005).

The results for the size based portfolios are shown in Table 1, and as expected the smallest size portfolio S-1 is giving the highest monthly excess returns[7] amounting to 3.63%, whereas the S-5 the biggest size portfolio is giving the minimum monthly excess returns of 0.92%. The column with the caption firms shows the average number of firms within each portfolio, in column ZR the average zero-returns of the firms in each portfolio is given. It is obvious that level of illiquidity is intrinsically linked with size of the firms and therefore with the returns as well. This can be ascertained by the column size in Table 1, which are monotonically increasing.

[Insert Table 1 here]

In Table 2, the characteristics of volatility related portfolios are shown. Volatility of each stocks in monthly variance of the stocks returns, which is estimated for preceding months and then on the basis of it, stock returns in succeeding months are predicted. The portfolio construction mechanism is same as for size based portfolios. Accordingly V-1 is the collection of those stocks whose volatility is the minimum, whereas V-2, V-3, V-4 and V-5 are the portfolios of those stocks whose volatility is monotonically increasing. The results in Table 2 are quite expected, the most volatile portfolio is the one giving the maximum monthly return of 4.16%, whereas V-1, the least volatile portfolio is giving the monthly returns of -0.04%. The column ZR and volatility shows that illiquidity is not monotonically linked, either with returns or with volatility. Therefore the construction of volatility related portfolios is unlike size related portfolios is independent of level of illiquidity. Nevertheless both of these portfolios may be exposed to market-wide illiquidity risk, which is a systematic dimension of illiquidity effect.

[Insert Table 2 here]

[7] The risk free rates for Pakistan is taken from State Bank of Pakistan.

4. Methodology

To price the return structure of the above constructed test assets, the unconditional version of LCAMP proposed by Acharya and Pedersen (2005) and also used in Lee (2011) is presented as under,

$$E(R_i - R_f) = E(C_i) + \lambda\ \beta_{i,1} + \lambda\beta_{i,2} - \lambda\beta_{i,3} - \lambda\beta_{i,4}, \tag{2}$$

in above model , $E(C_i)$ is expected level of illiquidity of test assets, whereas other four indicated betas are estimated using following relationships,

$$\beta_{i,1} = Cov(R_{i,t}, C_{m,t}) / Var(R_{m,t} - C_{m,t}), \tag{3}$$

$$\beta_{i,2} = Cov(R_{i,t}, C_{m,t}) / Var(R_{m,t} - C_{m,t}), \tag{4}$$

$$\beta_{i,3} = Cov(R_{i,t}, C_{m,t}) / Var(R_{m,t} - C_{m,t}), \tag{5}$$

$$\beta_{i,4} = Cov(C_{i,t}, R_{m,t}) / Var(R_{m,t} - C_{m,t}). \tag{6}$$

The equation (3) represents the usual market betas. In equation (4) commonality in illiquidity (studied by Chordia et al. (2002)) related betas is shown, which sees the impact of covariance of asset illiquidity, shown as $C_{i,t}$ and market illiquidity , shown as $C_{m,t}$ over its returns, which is positive. As the asset that becomes illiquid when market is illiquid requires some compensation for investors to hold such assets which are not hedged against market-wide illiquidity risk. In equation (5), the illiquidity risk that capture flight to liquidity effect is shown, an asset whose return increases when market illiquidity increases provide the cushion to the investors when illiquidity at market level increases. Resultantly it is priced negatively as shown in equation (2), the studies like Amihud (2002), Pastor and Stambaugh (2003), Bekaert et al. (2007) and others analyzed the pricing implication of this dimension of illiquidity risk. Lastly, in equation (6), the impact of market-wide returns over asset's illiquidity is shown. When market returns are depressed and the illiquidity of the stock reduces then such characteristic of an asset provide an ease to trade in adverse times. Resultantly, this illiquidity beta is priced negatively, showing higher demand of such assets. Studies like Acharya and Pedersen (2005) and Lee (2011) find that this dimension of illiquidity risk is the most important for the US market and for global markets.

4.1 Estimation of Betas and their Characteristics

Illiquidity for market portfolio and for other five size and volatility related portfolios each, is estimated using equation (2). Following literature Acharya and Pedersen (2005), Lee (2011) and Sadka (2005), instead of working with monthly illiquidity series directly, innovation in illiquidity series are used. As the illiquidity series are generally highly auto correlated, for instance this autocorrelation is 0.84 for the aggregate zero returns series of market portfolio. Similarly for size related five portfolios the auto correlation coefficient is from the 0.54 to .73, and for volatility related five portfolios the range is 0.67 to 0.81. To get the innovation in zero returns ARMA (1, 1)[8] is used. The innovations from this model are collected for monthly illiquidity of market and for other ten portfolios. Now the autocorrelations are significantly dropped, for market-wide illiquidity this correlation is now 0.02 and it is insignificant. Similarly the innovation in illiquidity series for size related portfolio is now within the range of 0.002 to 0.08 and for volatility portfolio the innovation in illiquidity series lies in the range of 0.03 to 0.18.

[8] By using AR(2) model the innovation is illiquidity series the autocorrelation are higher, however the overall content of the results presented in coming section of empirical analysis remain the same.

Using these innovation in illiquidity series and excess return series for test portfolios and market portfolio the betas shown in equation (3),(4),(5) and (6) are calculated. The characteristics of these betas are shown in Table 1 and 2.

Table 1 summarizes the betas related relationship of size portfolios, the β_1 shows the market beta associated with each portfolio, the exposure of smaller size S-1 portfolios is higher (1.121) to market risk in comparison to bigger size portfolio S-5 (0.814). The commonality in liquidity β_2 is showing counterintuitive exposures. As it is generally expected that illiquidity of smaller portfolio increases more with the increase in market-wide illiquidity. However its value for S-1 is 0.911 and for S-5 it is 1.251. Another form of illiquidity risk is β_3, now there is significant dispersion, for instance β_3 for S-1 is -0.340 and for S-5 it is -0.214, the higher time series negative relationship indicates as in Amihud (2002), that the returns of the smallest portfolio decreased the most when market's illiquidity increases. Lastly the β_4 also varies monotonically in relation with size, such that S-1 have the highest negative exposure of -0.271, whereas S-5 has the minimum exposure of -0.107. Intuitively when returns on market portfolio decreases, then innovation in zero returns increases the most for S-1, that is, under depressed market conditions the smaller stocks becomes more illiquid .

Table 2 summarizes the betas related relationship with volatility portfolios, the market beta β_1 shows significant variation with volatility related portfolios returns, the commonality in liquidity manifested through β_2 is not directly linked, whereas with, β_3 and β_4 this linkage is visible particularly the β_3 is quite monotonically linked with the returns on volatility portfolios. These characteristics of betas indicate that market risk β_1 and illiquidity related risk β_3 for both, size and volatility related portfolios capture the variation in returns over the time.

4.2 Stock-Based Test Assets.

As the number of stocks traded in KSE are not that high to construct larger number of portfolios for cross-sectional analysis. Therefore we use stocks as our test assets, this procedure has multiple advantages. First, a lot of information of stock returns is wasted when analysis are conducted at portfolio's level as individual stock returns are averaged out. Especially for small underdeveloped markets when return variation is quite high the potential loss of information is higher. Second, the number of portfolios are usually small and when coefficient of interest are estimated the degree of freedom are reduced. To circumvent this, using the stocks for cross-sectional analysis significantly improves the estimation procedure. There is also a drawback associated with stock-based analysis, that is, the model related risk, the betas are estimated with large estimation errors. The usual procedure to handle error in variable problem is that betas are estimated at portfolios level.

For example for the portfolio of smaller size stocks S-1, the respective betas are estimated using equation (2),(3),(4) & (5), afterwards each stock in S-1 is allocated the respective betas of the that portfolio. The same procedure is adopted for other portfolios. However the level of illiquidity for each stock is its zero-returns in preceding months. This procedure of allocating the level of illiquidity of stock and its betas risk break the strong correlation among them which is, handful in disentangling the effect of level of illiquidity from beta risks. As level of illiquidity changes for each stock over the time whereas, betas remain same. In Table 3, under the column of ZR, the correlation between level of illiquidity with betas risks is shown for size and volatility related portfolios. These correlations are significantly reduced at stock level in comparison to the correlations estimated at portfolio level[9]. Nevertheless, the betas related correlations are unaffected, as these are the same for stocks and portfolios. These correlations are quite high

[9] The correlations at portfolios level can be provided upon request.

especially between β_1 and β_3 for both types of stocks, either size related or volatility. For size related stocks this correlation is -.885 and for volatility stocks it is -0.972.

[Insert Table3 here]

4.3 Empirical Analysis.

The following testable version of LCAPM proposed by Acharya and Pedersen (2005) is estimated using Fama and Macbeth (1973) cross-sectional procedure, to analyze the explanatory power of local level of illiquidity, illiquidity risk and market risk,

$$E(R_i - R_f) = \alpha E(C_i) + \lambda_1\beta_{i,1} + \lambda_2\beta_{i,2} - \lambda_3\beta_{i,3} - \lambda_4\beta_{i,4} \tag{7}$$

to test the above model, owing to high correlation among betas as shown in Table 3, the betas are not included within any regression except for the last one, and that is to highlight the issues of muticollinearity. In Table 4, M1 is a model in which only the level of illiquidity is included to find its impact on the pricing of size related portfolios. In M2 level of illiquidity along with the market beta is tested to see the total impact of these two risk factors. In M3, M4 and M5, the level of illiquidity with illiquidity related betas are separately tested to see among three different illiquidity related risk candidates, which one is the most relevant. Lastly, in M5 the model is tested with the inclusion of level of illiquidity, market risk and constituent illiquidity related risks. In panel-B, the same procedure is repeated for volatility related portfolios.

In Table 4, the coefficient on the level of illiquidity for the size related portfolio is positive with the value 0.034 and associated t-stat of 3.62. Using this coefficient and following relationship,

$$\alpha \times \{E(C_1) - E(C_5)\}$$

where $E(C_1)$ is expected illiquidity on the smallest portfolio S-1, and $E(C_5)$ is expected illiquidity on the S-5 portfolio, these values are given in Table 1. Using the coefficient of expected illiquidity 0.034 and the average illiquidity of the respective portfolios, the monthly return differential 0.034x (0.49-0.21) =0.00952 is explained by the level of illiquidity. Whereas, the actual monthly return dispersion between these two portfolios is 0.0271 as given in Table 1, resultantly the total of 35% variation is return is explained by the level of illiquidity. In M2 model, the total explanation of return through level of illiquidity and market risk can be gauged by the following relationship,

$$\alpha \times \{E(C_1) - E(C_5)\} + \lambda_1 \times (\beta_{i,1} - \beta_{i,5}) \tag{8}$$

the coefficient on level of illiquidity α is now 0.0197 and price of market risk λ_1 is 0.0601, both are positive and statistically significant. Using the differential between expected illiquidity and market risk between the portfolio S-1 and S-5 given in Table 1, the relation (8) predicts this differential 0.0197x(0.49-0.21)+0.0601x(1.121-0.831) to be 0.0229. That is, M2 explains 85% of returns differential. Of this, level of illiquidity explains 20% and market risk explain 65%. The M3 model the price commonality is liquidity risk is counter intuitive, using M4 and relationship (8), with price of risk associated with flight to liquidity effect of -0.107 , level of illiquidity coefficient of 0.0222 as shown in Table 4 and respective illiquidity risk given in Table 1 under β_3 , the predicted monthly premium is 0.0197. That is 73% of the returns differential is explained by level and risk associated with illiquidity effect. Of which, contribution of level of illiquidity premium is 23% and of illiquidity risk is 50%. Similarly, the premium explained by model M4, using the illiquidity risk β_4 and associated price of risk and relationship (8) is 0.01889. As such the most economically meaningful illiquidity risk is β_3 in the context of KSE, this result is different than the results in Acharya and Pedersen (2005) and Lee (2011), as in their studies β_4 is the most significant illiquidity related risk. It seems that significance of indicated illiquidity risk is country specific. Lastly, in M6 all of the constituent risk factor of equation (7) are estimated, although only level of illiquidity and β_3 have the theoretically tenable sings and significance but nevertheless, these results are affected by multicollineraity, for instance the magnitude of price of risk associated with flight to liquidity effect is increased but its statistical significance is reduced in comparison to model M4.

[Insert Table 4 here]

In Panel B, Table 4 the results for volatility related portfolios are summarized. Here we find the negative coefficient on the level of illiquidity, as the stocks related with their volatility show no monotonic relationship with the level of illiquidity as shown in Table 2, column ZR. Therefore this result means, level of illiquidity for the volatility related stocks is not economically important. Using the output of the model M2 in Table 4, the corresponding variables given in Table 2 and equation (8), the predicted premium is 0.04255. Whereas, the actual return dispersion between V-5 and V-1, given in Table 2 is .04200 on monthly basis. Similarly using the output of M4, this predicted premium is 0.0438. Generally, all of the excess return is predicted either by the market risk or by the illiquidity risk β_3. The other two illiquidity risks are not important for the pricing of volatility related stocks. These results also hint, that even if level of illiquidity is not linked with the stock returns, the market-wide illiquidity risk is still significant part of the pricing of such stocks.

4.4 Diversification

There is a series of papers (Bekaert and Greet (1995), Bekaert, Geert, and Campbell R. Harvey (1995), (1997), (2000) and others), in which diversification of portfolio risk by the inclusion of the stocks traded in emerging markets is discussed. The volatility of higher returns and higher illiquidity in such markets is compensated at local level, as is the case with KSE analyzed in previous sections. However, for international investors this volatility and illiquidity, is not translated into risk till the time it results into higher correlation with the risk factors against which they aspire to hedge their portfolios returns, or demand the compensation for being exposed to. To elaborate this, we take the example of the US investor and assume that the three factors, market, size and value of Fama and French (1993) are the true source of risk. Now if the size and volatility related portfolios in KSE are equally exposed to such factors as are the returns on comparable portfolios in the US market, then no hedging benefits can be achieved by the US investor. Naturally, we concentrate on alphas of three factor model[10] (constructed for the US market) for five size and volatility related portfolios (test assets based on KSE stocks). The returns in KSE market are converted in US dollars and to get excess returns the risk free rates given in Fama and French website are used.

In Table 5, the estimated out-put of the three factor model of Fama and French (1993) is given. There is statistically significant exposure of the returns of these KSE size and volatility related portfolios on risk factors for the US market. For instance, the price of risk λ_{smb} associated with SMB factor for the least capitalized portfolio S-1 is the most significant, but for the largest capitalized portfolio S-5 both the economic and statistical significance of size factor is reduced. Same also hold for V-1 and V-5. Further, the exposure of market returns in the US market is also

[10] The three factors for the US market are taken from the Fama and French data library.

significant for each portfolio. Nevertheless, the alphas of the model, indicating the excess returns not explained by the model are linearly related with the excess returns on these portfolios, that is, higher the returns higher the alphas.

The monthly excess dollar return on the portfolio S-1 and S-5 are 3.68% and 1.05% in the KSE. Whereas, 2.87% and 0.44% monthly returns of these portfolios are not explained as shown in Table 5. Similarly for portfolio V-5, the excess returns are 4.27%, whereas 3.25% are not explained. On the other hand, the excess returns on size related ten portfolios for the US market are well explained by the three factor model[11]. For instance, the alpha on S-1 and S-10, for the US based sized related portfolios are -0.17% and -0.12%, that is after accounting for the risks there is no excess returns on these portfolios remain available.

[Insert Table 5 here]

The above analysis indicate that an international investor by investing in stocks traded in KSE can get higher returns and at the same time reduce the risk. As returns on KSE are not that correlated with the risk factors that are quite pertinent for international investor. Nevertheless, these results are based on the extreme assumption of total liberalization. However, as indicated in previous research that markets like KSE are neither fully integrated nor segmented. Although the official liberalization date[12] for Pakistan is February 1991, nevertheless all stocks still remain practically inaccessible to foreigners. To proxy for the investable stocks we downloaded from DataStream, S&P/IFCG Extended Frontier 150 Index for Pakistan, which include the most liquid and larger

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[11] The ten equal weighted size related portfolios and respective three risk factors are downloaded by Fama and French data library, the detail results on the estimation of this model are available upon request.
[12] These liberalization dates for different markets are given in Bekaert and Harvey (2000).

capitalized firms traded in KSE. The index is available from November 2008 onwards, the average number of stocks from KSE are 17. If this portfolio is held by the US investor then the alpha from three factor Fama and French (1993) model is 1.64% on monthly basis. On the other hand the alpha is -0.0041%, that is, practically non-existent when the CAPM implying local market risk is used. Therefore, even for the most investable stocks in KSE the local risk matters the most, the risks for instance for the US investor do not count. These results indicate the diversification opportunities are available for foreign investors by holding the stocks from emerging/frontier markets in their portfolios.

5. Conclusion

The anomalous returns using the publically available information are reported extensively across different markets as a challenge to efficient market hypothesis. Although these returns in terms of magnitude are not that high in the developed markets, but still managed to attract a lot of empirical inquisition. For instance for the US market the return differential between the least and the largest capitalized equally weighted portfolio is 4.69% on annual basis for the period of 1993-2015, same holds true for volatility, book-to-market, operating profits and investment etc. related firms characteristics. On the other hand for the emerging/frontier markets like KSE, the annual returns based on size and volatility based strategy are 31.57% and 50.43% on annual basis. The lesser focus is given in literature to rationalize these higher returns within the framework of efficient market hypothesis. This study fulfils this gap and analyzed that higher returns is emerging/frontier markets is not a manifestation of inefficiency of the market. As within the pricing model of LCAPM proposed by Acharya and Pedersen (2005), almost all of the extra-ordinary returns are linked with the local risk premiums that investors demand in terms of effect of illiquidity and market risk to which investors are exposed.

Although these local risks are very important for pricing of the stocks in KSE, but their return structure remain isolated to international risk factors proxy by the market, size and value factors for the US market. Such that for size and volatility based KSE portfolios the return differential of 2.43% and 3.60% between the extreme portfolios is not explained. These results just do not confine for such stocks which are least capitalized and illiquid and therefore inaccessible to foreign investors. Even for highly capitalized and liquid stocks in KSE that constitute a part of S&P/IFCG extended 150 index, the monthly returns of 1.64% is not explained. Whereas, the returns on these stocks are totally rationalized within the simple CAPM using the local market risk factor. This indicates that the opportunities for portfolio diversification for international investors are quite real.

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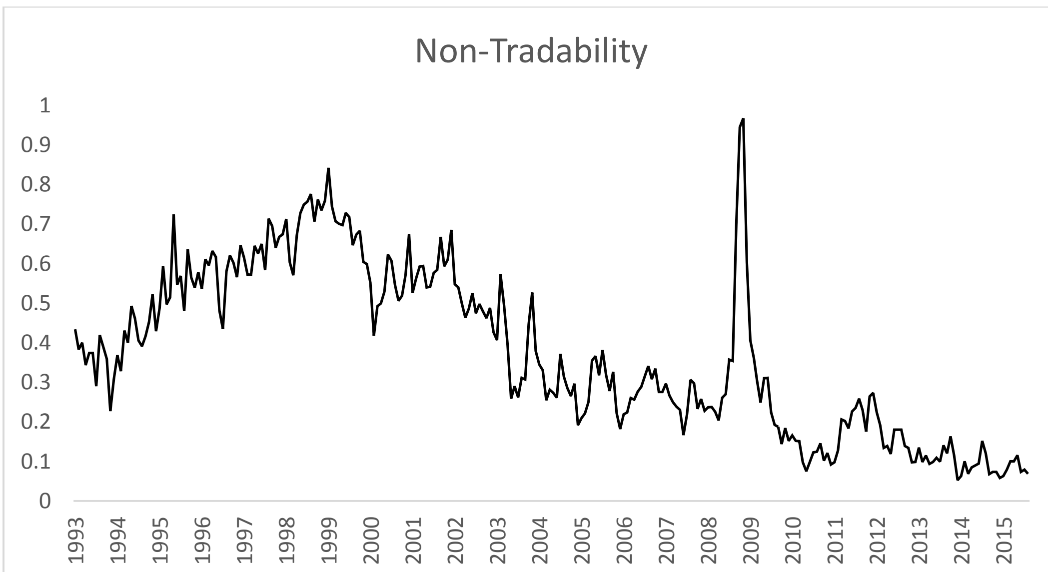

Figure 1: Average number of firms trading for maximum of three days in a month from 1993-2015.

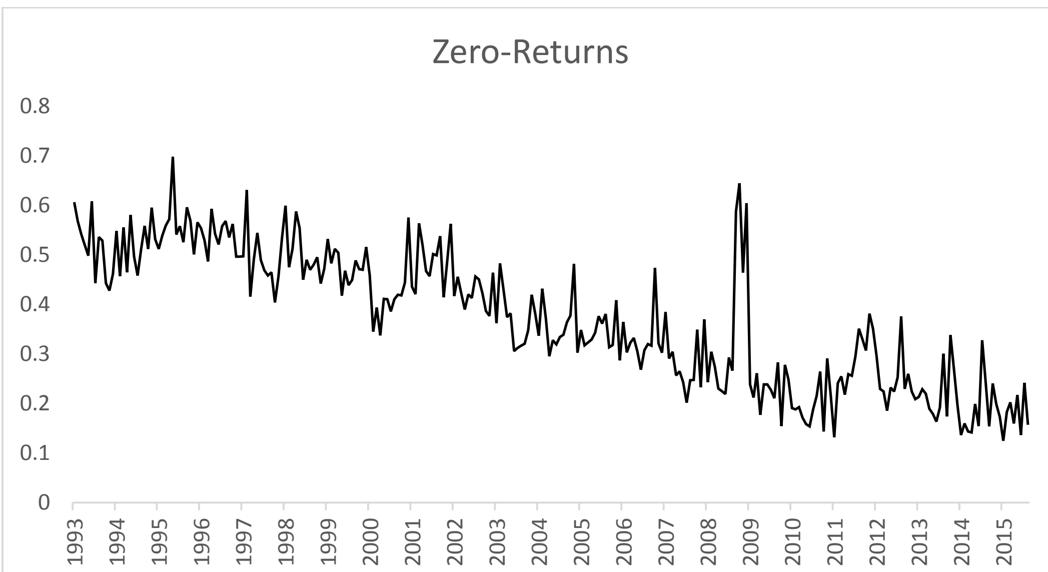

Figure 2: Monthly average of zero-returns of the firms traded in KSE for the period of 1993-2015.

Table 1: Size Portfolios Related Characteristic.

This Table summarizes the characteristics of size related portfolios. S-1 is the portfolio which is comprised of approximately 20% of least capitalized stocks in KSE, Pakistan. Portfolios like S-2, S-3, S-4 and S-5 are those firms whose market capitalization is increasing monotonically by 20% for each portfolio, such that S-5 is the portfolio composed of approximately 20% of highly capitalized stocks. The monthly returns of these portfolios for the period 1993-2015 are shown under the heading of returns, firms' shows average number of stocks in each portfolio. ZR is monthly average of zero returns of the firms whereas, size is the average market capitalization of these firms which is product of number of shares outstanding and end of month prices, size is shown in Pak Rupees (in billion). The market beta and illiquidity related betas for these portfolios are shown as β_1, β_2, β_3 and β_4 and estimated using equation (2),(3),(4) &(5).

Portfolios	Returns	Firms	ZR	Size	β_1	β_2	β_3	β_4
S-1	3.63%	23	49%	0.09	1.121	0.911	-0.340	-0.271
S-2	2.40%	24	41%	0.49	0.935	1.089	-0.325	-0.247
S-3	1.65%	25	35%	1.48	0.921	1.034	-0.291	-0.233
S-4	1.57%	26	28%	4.14	0.831	0.967	-0.230	-0.138
S-5	0.92%	27	21%	38.76	0.814	1.251	-0.214	-0.107

Table 2: Idiosyncratic Volatility Portfolios Related Characteristic.

This Table summarizes the characteristics of size related portfolios. V-1 is the portfolio which is comprised of approximately 20% of those stocks whose volatility is the least in KSE, Pakistan. Portfolios such as V-2, V-3, V4 and V-5 are those firms whose market volatility is increasing monotonically each by 20%, such that V-5 is the portfolio composed of approximately 20% of highly volatile stocks. The monthly returns of these portfolios for the period 1993-2015 are shown under the heading of returns, firms' shows average number of stocks in each portfolio. ZR is monthly average of zero returns of the firms included in each portfolio whereas, Volatility is the average volatility of these firms .The market beta and illiquidity related betas for these portfolios are shown as β_1, β_2, β_3 and β_4 and estimated using equation (2),(3),(4) &(5).

Portfolios	Returns	Firms	ZR	Volatility	β_1	β_2	β_3	β_4
V-1	-0.04%	23	33%	2.01%	0.498	1.033	-0.141	-0.247
V-2	0.92%	25	31%	2.49%	0.698	1.088	-0.226	-0.206
V-3	1.69%	25	31%	2.90%	0.872	0.945	-0.247	-0.066
V-4	2.26%	25	33%	3.84%	1.121	1.193	-0.283	-0.175
V-5	4.16%	24	42%	6.85%	1.419	1.066	-0.342	-0.234

Table 3: Correlation Structure

This Table presents the correlation among model constituent variables for the size and volatility related portfolios. In Panel A, the correlation between level of illiquidity shown as ZR, and market beta β_1 and other illiquidity related betas β_2 , β_3 and β_4 are shown. The ZR is estimated as previous month's number of zero returns for each stock whereas, each stock is allocated its portfolio related beta. In Panel B, the same correlation structure is shown for volatility related portfolios.

Panel A:	ZR	β_1	β_2	β_3	β_4
ZR	1				
β_1	0.3751	1			
β_2	-0.2699	-0.6338	1		
β_3	-0.3768	-0.8848	0.5383	1	
β_4	-0.3690	-0.8631	0.5872	0.9866	1

Panel B:	ZR	β_1	β_2	β_3	β_4
ZR	1				
β_1	0.1513	1			
β_2	0.0141	0.3127	1		
β_3	-0.1400	-0.9723	-0.2870	1	
β_4	-0.0641	0.0197	-0.4615	-0.1182	1

Table 4: Stock based analysis for size and volatility based portfolios using Fama-MacBeth regressions.

The following Table presents the estimation of the Acharya and Pedersen (2005) Liquidity Adjusted CAPM,

$$E(R_i - R_f) = \alpha E(C_i) + \lambda_1 \beta_{i,1} + \lambda_2 \beta_{i,2} - \lambda_3 \beta_{i,3} - \lambda_4 \beta_{i,4}$$

The tests assets are the stocks which are grouped into five portfolios based upon their previous month's size and volatility. Subsequently, each stock is assigned market and illiquidity related betas of the portfolio to which that stock belongs. These betas are calculated using equation (3),(4),(5) and (6). The expected illiquidity ZR is stock's previous month average zero returns. Panel A, represents the estimated coefficients of the test assets on expected illiquidity and model related risk, the t-stat are shown below the coefficients in parenthesis. Panel B, repeats the same procedure for volatility based portfolios. These results are based for the period of 1993-2015.

	M1	M2	M3	M4	M5	M6
ZR	0.034	0.0197	0.0283	0.0222	0.0237	0.0228
	(3.62)	(2.10)	(3.13)	(2.30)	(2.43)	(2.41)
β_1		0.0601				0.0039
		(3.08)				(0.10)
β_2			-0.0368			-0.0476
			(-2.77)			(-1.76)
β_3				-0.107		-0.336
				(-2.85)		(-2.17)
β_4					-0.0747	0.199
					(-2.65)	(1.75)
Constant	0.0180	-0.0314	0.0589	-0.0066	0.0078	0.014
	(2.40)	(-1.89)	(3.36)	(-0.66)	(1.05)	(0.31)

Panel B: Volatility Based Portfolios

	M1	M2	M3	M4	M5	M6
ZR	-0.0502	-0.0635	-0.0487	-0.0621	-0.0497	-0.0621
	(-5.99)	(-7.02)	(-5.79)	(-6.85)	(-6.02)	(-6.95)
β_1		0.0524				0.0493
		(6.10)				(3.55)
β_2			0.0353			-0.0512
			(2.51)			(-2.74)
β_3				-0.246		-0.0357
				(-6.14)		(-0.69)
β_4					-0.0272	-0.0585
					(-1.44)	(-2.38)
Constant	0.0419	-0.00367	0.00395	-0.0170	0.0366	0.0331
	(5.45)	(-0.54)	(0.28)	(-2.17)	(4.21)	(2.32)

Table 5: Relationship between local returns with international risk factors.

This Tables presents the results of Fama and French (1993) three factor model, by using the risk factors for the US market which are excess market return MR, size factor SMB, value factor HML.

$$E(R_i - R_f)_t = \alpha_{i,t} + \lambda_{i,m}(MR - R_F)_t + \lambda_{i,smb}(SMB)_t + \lambda_{i,hml}(HML)_t + \varepsilon_{i,t}$$

The test assets are excess return on the size and volatility based portfolios five portfolios for the KSE, Pakistan. The returns are denominated in US$, the time period of the analysis is 1993-2015. The t-stats for each coefficient is presented below in prentices, the last column shows the R^2 of each model and adjusted R^2 is presented below in prentices.

Portfolios	Constant	MR	SMB	HML	R^2
S-1	0.0287	0.415	0.929	0.557	0.084
	(3.56)	(2.56)	(3.59)	(2.12)	(0.074)
S-2	0.0182	0.454	0.557	0.216	0.086
	(2.73)	(3.40)	(2.51)	(1.00)	(0.076)
S-3	0.0114	0.407	0.283	0.174	0.056
	(1.80)	(3.19)	(1.39)	(0.84)	(0.045)
S-4	0.0127	0.334	0.308	0.100	0.059
	(2.11)	(2.75)	(1.59)	(0.05)	(0.049)
S-5	0.0044	0.446	0.333	0.167	0.075
	(0.72)	(3.63)	(1.70)	(0.84)	(0.065)
Panel B: Volatility Portfolios					
Portfolios	Constant	MR	SMB	HML	R^2
V-1	-0.0035	0.286	0.187	0.082	0.070
	(-0.90)	(3.49)	(1.51)	(0.65)	(0.059)
V-2	0.0059	0.294	0.289	0.029	0.060
	(1.15)	(2.71)	(1.76)	(0.17)	(0.049)
V-3	0.012	0.417	0.289	0.074	0.060
	(1.86)	(3.06)	(1.41)	(0.35)	(0.049)
V-4	0.0177	0.446	0.497	0.087	0.069
	(2.34)	(2.94)	(2.05)	(0.35)	(0.059)
V-5	0.0325	0.622	0.725	0.594	0.066
	(3.28)	(3.13)	(2.28)	(1.84)	(0.055)